

Worked Example: Belief Propagation for a 4×4 Sudoku

1 Problem setup: 4×4 Sudoku as a pairwise CSP

We consider a 4×4 Sudoku with alphabet $\mathcal{X} = \{1, 2, 3, 4\}$. The variable nodes are indexed by

$$i = (r, c) \in \{1, 2, 3, 4\}^2,$$

and the variable at index i is written as x_i . This simplifies notation by allowing us to treat x_i with $i = (r, c)$ and $x_{r,c}$ as the same variable. Constraints:

- Each row is a permutation of $\{1, 2, 3, 4\}$ (equivalently, all-different).
- Each column is a permutation of $\{1, 2, 3, 4\}$ (equivalently, all-different).
- Each 2×2 sub-block is a permutation of $\{1, 2, 3, 4\}$ (equivalently, all-different).

We implement these all-different constraints via *pairwise not-equal factors*

$$f_{\neq}(x_i, x_j) = \mathbf{1}[x_i \neq x_j],$$

added on every pair of cells that share a row, column, or 2×2 block.

Unary factors (givens). For variable $x_{(r,c)}$ fixed to value $v \in \{1, 2, 3, 4\}$, we use a unary factor

$$f_i(x_i) = \mathbf{1}[x_i = v].$$

For a blank cell, we use *no pruning* for this worked example:

$$f_i(x_i) \equiv 1, \quad x_i \in \{1, 2, 3, 4\}.$$

2 The test puzzle (chosen to converge quickly under BP)

We use the following puzzle (0 denotes blank):

1	0	0	4
0	4	0	2
0	0	4	0
4	0	0	1

The (unique) solution is

1	2	3	4
3	4	1	2
2	1	4	3
4	3	2	1

3 Loopy BP update equations for pairwise not-equal factors

Index the 16 variables by their coordinates $i = (r, c) \in \{1, 2, 3, 4\}^2$ where $x_i \equiv x_{r,c}$. Two variables are neighbors if they share a row, column, or 2×2 block. In this example, each variable is involved in 3 permutation constraint factors (row, column, and block). But, to reduce complexity, each permutation factor is decomposed into 6 pairwise not-equal factors. Thus, in the graph used by BP, each variable node has degree 7.

We run BP on the *pairwise* factor graph, so we can simplify things by treating all messages as variable-to-variable messages on directed edges. Let $m_{i \rightarrow j}(x)$ be the message from variable i to variable j (a vector in $\mathbb{R}_+^4$ indexed by $x \in \{1, 2, 3, 4\}$), where $i, j \in \{1, 2, 3, 4\}^2$. For example, if $i = (r, c)$ and $j = (r', c')$, we write

$$m_{i \rightarrow j} \equiv m_{(r,c) \rightarrow (r',c')}.$$

Factor-to-variable shortcut. To create variable-to-variable messages, we collapse the variable-to-factor $i \rightarrow a$ and factor-to-variable $a \rightarrow j$ messages into a single $i \rightarrow j$ message. For a not-equal factor $a = (i, j)$ where $i, j \in \{1, 2, 3, 4\}^2$ share a factor, the factor-to-variable update is

$$\hat{m}_{a \rightarrow i}(x_i) = \sum_{x_j \in \{1,2,3,4\}} \mathbf{1}[x_i \neq x_j] m_{j \rightarrow a}(x_j) = -m_{j \rightarrow a}(x_i) + \sum_{x_j \in \{1,2,3,4\}} m_{j \rightarrow a}(x_j).$$

If $m_{j \rightarrow a}(x)$ is normalized to sum to 1, then

$$\boxed{\hat{m}_{a \rightarrow i}(x) = 1 - m_{j \rightarrow a}(x)}.$$

Since the factor graph is pairwise, we can treat $m_{j \rightarrow a}$ with $a = (i, j)$ as the message from j to i .

Variable-to-variable update. Let $N(i)$ be the neighbor set of variable i in the pairwise factor graph (i.e., the set of variables sharing a not-equal factor with variable i). Then

$$\boxed{m_{i \rightarrow j}(x) \propto f_i(x) \prod_{k \in N(i) \setminus \{j\}} \hat{m}_{k \rightarrow i}(x) = f_i(x) \prod_{k \in N(i) \setminus \{j\}} (1 - m_{k \rightarrow i}(x))}$$

followed by normalization so that $\sum_{x \in \{1,2,3,4\}} m_{i \rightarrow j}(x) = 1$.

Beliefs. After each round, define the (approximate) marginal for i by

$$\boxed{b_i(x) \propto f_i(x) \prod_{k \in N(i)} (1 - m_{k \rightarrow i}(x)), \quad \sum_{x \in \{1,2,3,4\}} b_i(x) = 1.}$$

Notation for vectors. We write probability vectors in the fixed digit order $(1, 2, 3, 4)$, e.g.

$$m_{i \rightarrow j} = [p_1, p_2, p_3, p_4].$$

4 Round 0 initialization

For givens, messages are one-hot. For blanks, we initialize uniformly.

- If $x_{r,c}$ is given as v , then for every neighbor j ,

$$m_{(r,c) \rightarrow j}^{(0)} = \mathbf{e}_v, \quad \text{where } \mathbf{e}_1 = [1, 0, 0, 0], \mathbf{e}_2 = [0, 1, 0, 0], \mathbf{e}_3 = [0, 0, 1, 0], \mathbf{e}_4 = [0, 0, 0, 1].$$

- If $x_{r,c}$ is blank, then for every neighbor j ,

$$m_{(r,c) \rightarrow j}^{(0)} = [\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}].$$

5 Neighbor sets needed for the worked computations

We will work out computations for three representative cells. Their neighbor sets are:

$$N(x_{(1,2)}) = \{x_{(1,1)}, x_{(1,3)}, x_{(1,4)}, x_{(2,1)}, x_{(2,2)}, x_{(3,2)}, x_{(4,2)}\},$$

$$N(x_{(3,2)}) = \{x_{(1,2)}, x_{(2,2)}, x_{(3,1)}, x_{(3,3)}, x_{(3,4)}, x_{(4,1)}, x_{(4,2)}\},$$

$$N(x_{(4,2)}) = \{x_{(1,2)}, x_{(2,2)}, x_{(3,2)}, x_{(3,1)}, x_{(4,1)}, x_{(4,3)}, x_{(4,4)}\}.$$

6 Round 1: compute messages and beliefs

In Round 1 we use the update rule with incoming messages from Round 0.

6.1 Some immediate outgoing messages in Round 1

Because many neighbors are givens, several cells become effectively forced after the first sweep. We list the key Round-1 outgoing messages that will be used below (all are derived directly from Round 0 messages):

- $x_{(1,3)}$ shares row with $x_{(1,1)} = 1$, shares row with $x_{(1,4)} = 4$, shares block with $x_{(2,4)} = 2$, and shares column with $x_{(3,3)} = 4$; hence digits $\{1, 2, 4\}$ are eliminated and only 3 remains:

$$m_{(1,3) \rightarrow j}^{(1)} = \mathbf{e}_3 \quad \text{for every neighbor } j.$$

- $x_{(2,1)}$ shares column with $x_{(1,1)} = 1$ and $x_{(4,1)} = 4$, and shares row with $x_{(2,2)} = 4$ and $x_{(2,4)} = 2$; hence only 3 remains:

$$m_{(2,1) \rightarrow j}^{(1)} = \mathbf{e}_3 \quad \text{for every neighbor } j.$$

- $x_{(3,4)}$ shares column with $x_{(1,4)} = 4$, $x_{(2,4)} = 2$, $x_{(4,4)} = 1$, hence only 3 remains:

$$m_{(3,4) \rightarrow j}^{(1)} = \mathbf{e}_3 \quad \text{for every neighbor } j.$$

- $x_{(3,2)}$ sees three neighbors equal to 4 (namely $x_{(2,2)}, x_{(3,3)}, x_{(4,1)}$), so digit 4 is eliminated but $\{1, 2, 3\}$ remain symmetric under Round 0:

$$m_{(3,2) \rightarrow j}^{(1)} = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0] \quad \text{for every neighbor } j.$$

- $x_{(4,2)}$ sees neighbors $x_{(4,4)} = 1$ and $x_{(2,2)} = 4$ and $x_{(4,1)} = 4$, eliminating digits 1 and 4, leaving $\{2, 3\}$ symmetric under Round 0:

$$m_{(4,2) \rightarrow j}^{(1)} = [0, \frac{1}{2}, \frac{1}{2}, 0] \quad \text{for every neighbor } j.$$

6.2 Worked belief computation 1: $x_{(1,2)}$ after Round 1

We compute $b_{(1,2)}^{(1)}(x)$ using

$$b_{(1,2)}^{(1)}(x) \propto \phi_{(1,2)}(x) \prod_{k \in N((1,2))} (1 - m_{k \rightarrow (1,2)}^{(1)}(x)).$$

Here $\phi_{(1,2)} \equiv 1$ because $(1,2)$ is blank.

Incoming messages (in digit order $(1, 2, 3, 4)$). Here we write $m_{i \rightarrow j} \equiv m_{(r,c) \rightarrow (r',c')}$ with $i = (r, c)$ and $j = (r', c')$, e.g., $m_{(1,1) \rightarrow (1,2)} = m_{(1,1) \rightarrow (1,2)}$.

$$\begin{aligned} m_{(1,1) \rightarrow (1,2)}^{(1)} &= \mathbf{e}_1 = [1, 0, 0, 0], \\ m_{(1,3) \rightarrow (1,2)}^{(1)} &= \mathbf{e}_3 = [0, 0, 1, 0], \\ m_{(1,4) \rightarrow (1,2)}^{(1)} &= \mathbf{e}_4 = [0, 0, 0, 1], \\ m_{(2,1) \rightarrow (1,2)}^{(1)} &= \mathbf{e}_3 = [0, 0, 1, 0], \\ m_{(2,2) \rightarrow (1,2)}^{(1)} &= \mathbf{e}_4 = [0, 0, 0, 1], \\ m_{(3,2) \rightarrow (1,2)}^{(1)} &= [\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, 0], \\ m_{(4,2) \rightarrow (1,2)}^{(1)} &= [0, \tfrac{1}{2}, \tfrac{1}{2}, 0]. \end{aligned}$$

Thus the corresponding factor-to-variable (“hat”) messages are

$$\begin{aligned} 1 - \mathbf{e}_1 &= [0, 1, 1, 1], \quad 1 - \mathbf{e}_3 = [1, 1, 0, 1], \quad 1 - \mathbf{e}_4 = [1, 1, 1, 0], \\ 1 - [\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, 0] &= [\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}, 1], \quad 1 - [0, \tfrac{1}{2}, \tfrac{1}{2}, 0] = [1, \tfrac{1}{2}, \tfrac{1}{2}, 1]. \end{aligned}$$

Now multiply componentwise over the 7 neighbors:

digit x	1	2	3	4
$\prod_{k \in N((1,2))} (1 - m_{k \rightarrow (1,2)}^{(1)}(x))$	0	$\frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$	0	0

Hence $b_{(1,2)}^{(1)} = \mathbf{e}_2$:

$b_{(1,2)}^{(1)} = [0, 1, 0, 0]$

 $\Rightarrow (1, 2) = 2 \text{ (MAP).}$

6.3 Worked belief computation 2: $x_{(3,2)}$ after Round 1

From the Round-1 sweep, the incoming messages to $x_{(3,2)}$ are:

$$\begin{aligned} m_{(1,2) \rightarrow (3,2)}^{(1)} &= [0, \tfrac{1}{2}, \tfrac{1}{2}, 0], \quad m_{(2,2) \rightarrow (3,2)}^{(1)} = \mathbf{e}_4, \quad m_{(3,1) \rightarrow (3,2)}^{(1)} = [0, \tfrac{1}{2}, \tfrac{1}{2}, 0], \\ m_{(3,3) \rightarrow (3,2)}^{(1)} &= \mathbf{e}_4, \quad m_{(3,4) \rightarrow (3,2)}^{(1)} = \mathbf{e}_3, \quad m_{(4,1) \rightarrow (3,2)}^{(1)} = \mathbf{e}_4, \quad m_{(4,2) \rightarrow (3,2)}^{(1)} = [0, \tfrac{1}{2}, \tfrac{1}{2}, 0]. \end{aligned}$$

(Here $x_{(3,1)}$ and $x_{(1,2)}$ and $x_{(4,2)}$ are not givens; their Round-1 outgoing messages are symmetric on $\{2, 3\}$.)

Take hats:

$$1 - [0, \tfrac{1}{2}, \tfrac{1}{2}, 0] = [1, \tfrac{1}{2}, \tfrac{1}{2}, 1], \quad 1 - \mathbf{e}_4 = [1, 1, 1, 0], \quad 1 - \mathbf{e}_3 = [1, 1, 0, 1].$$

Multiply componentwise over the 7 neighbors:

$$\underbrace{[1, \frac{1}{2}, \frac{1}{2}, 1]^{\odot 3}}_{\text{three half-messages}} \odot \underbrace{[1, 1, 1, 0]^{\odot 3}}_{\text{three "4" givens}} \odot \underbrace{[1, 1, 0, 1]}_{\text{one "3" message}} = [1, \frac{1}{8}, 0, 0].$$

Since $f_{(3,2)} \equiv 1$ (blank), normalize:

$$\boxed{b_{(3,2)}^{(1)} = [\frac{8}{9}, \frac{1}{9}, 0, 0]}.$$

6.4 Worked belief computation 3: (4, 2) after Round 1

Incoming messages to $x_{(4,2)}$ in Round 1 include:

$$m_{(4,4) \rightarrow (4,2)}^{(1)} = \mathbf{e}_1, \quad m_{(2,2) \rightarrow (4,2)}^{(1)} = \mathbf{e}_4, \quad m_{(4,1) \rightarrow (4,2)}^{(1)} = \mathbf{e}_4,$$

$$m_{(1,2) \rightarrow (4,2)}^{(1)} = [0, \frac{1}{2}, \frac{1}{2}, 0], \quad m_{(3,2) \rightarrow (4,2)}^{(1)} = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0], \quad m_{(3,1) \rightarrow (4,2)}^{(1)} = [0, \frac{1}{2}, \frac{1}{2}, 0], \quad m_{(4,3) \rightarrow (4,2)}^{(1)} = [0, \frac{1}{2}, \frac{1}{2}, 0].$$

Thus hats are:

$$1 - \mathbf{e}_1 = [0, 1, 1, 1], \quad 1 - \mathbf{e}_4 = [1, 1, 1, 0], \quad 1 - [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0] = [\frac{2}{3}, \frac{2}{3}, \frac{2}{3}, 1], \quad 1 - [0, \frac{1}{2}, \frac{1}{2}, 0] = [1, \frac{1}{2}, \frac{1}{2}, 1].$$

Multiplying componentwise over all 7 neighbors forces digits 1 and 4 to zero and yields equal mass on 2 and 3:

$$b_{(4,2)}^{(1)} \propto [0, \frac{1}{12}, \frac{1}{12}, 0] \Rightarrow \boxed{b_{(4,2)}^{(1)} = [0, \frac{1}{2}, \frac{1}{2}, 0]}.$$

6.5 Belief table after Round 1 (all cells)

We now summarize the beliefs after Round 1. Each entry is $b_{(r,c)}^{(1)} = [p_1, p_2, p_3, p_4]$.

$[1, 0, 0, 0]$	$[0, 1, 0, 0]$	$[0, 0, 1, 0]$	$[0, 0, 0, 1]$
$[0, 0, 1, 0]$	$[0, 0, 0, 1]$	$[1, 0, 0, 0]$	$[0, 1, 0, 0]$
$[0, 1, 0, 0]$	$[\frac{8}{9}, \frac{1}{9}, 0, 0]$	$[0, 0, 0, 1]$	$[0, 0, 1, 0]$
$[0, 0, 0, 1]$	$[0, \frac{1}{2}, \frac{1}{2}, 0]$	$[0, 1, 0, 0]$	$[1, 0, 0, 0]$

Notice that after Round 1, the only remaining ambiguity is in $x_{(3,2)}$ and $x_{(4,2)}$, and even $x_{(3,2)}$ is already heavily polarized toward digit 1.

7 Round 2: resolve the remaining ambiguity

7.1 Belief of (3, 2) after Round 2

Using the updated incoming messages from Round 1, one obtains in Round 2 that $x_{(3,2)}$ is forced to digit 1:

$$\boxed{b_{(3,2)}^{(2)} = \mathbf{e}_1 = [1, 0, 0, 0]}.$$

(Heuristically: several neighbors now strongly force digit 2 or 3 elsewhere, and the not-equal factors eliminate them for $x_{(3,2)}$.)

7.2 Belief of $x_{(4,2)}$ after Round 2 (fully worked)

The incoming messages to $x_{(4,2)}$ in Round 2 are:

$$m_{(1,2) \rightarrow (4,2)}^{(2)} = \mathbf{e}_2, \quad m_{(2,2) \rightarrow (4,2)}^{(2)} = \mathbf{e}_4, \quad m_{(3,1) \rightarrow (4,2)}^{(2)} = \mathbf{e}_2, \quad m_{(3,2) \rightarrow (4,2)}^{(2)} = [\frac{4}{5}, \frac{1}{5}, 0, 0],$$

$$m_{(4,1) \rightarrow (4,2)}^{(2)} = \mathbf{e}_4, \quad m_{(4,3) \rightarrow (4,2)}^{(2)} = \mathbf{e}_2, \quad m_{(4,4) \rightarrow (4,2)}^{(2)} = \mathbf{e}_1.$$

Take hats:

$$1 - \mathbf{e}_2 = [1, 0, 1, 1], \quad 1 - \mathbf{e}_4 = [1, 1, 1, 0], \quad 1 - \mathbf{e}_1 = [0, 1, 1, 1],$$

$$1 - [\frac{4}{5}, \frac{1}{5}, 0, 0] = [\frac{1}{5}, \frac{4}{5}, 1, 1].$$

Now compute the product over all neighbors for each digit:

- Digit 1 is killed by $1 - \mathbf{e}_1$ (which has a 0 in position 1).
- Digit 2 is killed by $1 - \mathbf{e}_2$ (which has a 0 in position 2) from multiple neighbors.
- Digit 4 is killed by $1 - \mathbf{e}_4$ (which has a 0 in position 4) from multiple neighbors.
- Digit 3 survives all hats with value 1 in position 3.

Hence

$$\boxed{b_{(4,2)}^{(2)} = \mathbf{e}_3 = [0, 0, 1, 0]}.$$

7.3 Belief table after Round 2 (solution reached)

$[1, 0, 0, 0]$	$[0, 1, 0, 0]$	$[0, 0, 1, 0]$	$[0, 0, 0, 1]$
$[0, 0, 1, 0]$	$[0, 0, 0, 1]$	$[1, 0, 0, 0]$	$[0, 1, 0, 0]$
$[0, 1, 0, 0]$	$[1, 0, 0, 0]$	$[0, 0, 0, 1]$	$[0, 0, 1, 0]$
$[0, 0, 0, 1]$	$[0, 0, 1, 0]$	$[0, 1, 0, 0]$	$[1, 0, 0, 0]$

8 Final decoded grid (MAP from beliefs)

Taking $\arg \max_x b_{(r,c)}^{(2)}(x)$ yields:

1	2	3	4
3	4	1	2
2	1	4	3
4	3	2	1